

TABLE I
EM-18 RFID Module
Specifications

Heading	Specifications
Reading distance	6-10cm
Dimensions	40mm×20mm×8mm (L×H×W)
Frequency	125kHz
Compatible card codes	Manchester 64-bit, modulus 64 encoding
Current rating	35mA (max.)
Operating voltage	4.6V-5.4V DC

TABLE II
Test Points

Test point	Details
TP0	0V
TP1	12V

for programming the on-board microcontroller, power jack, an ICSP header and a reset button. It is operated with a 16MHz crystal oscillator and contains everything needed to support the microcontroller. It is very easy to use as the user simply needs to connect it to a computer with a USB cable or power it with an AC-to-DC adaptor or battery to get started. The microcontroller on the board is programmed using Arduino programming language and Arduino development environment.

Pin 0 (RX) of Board1 is connected to pin 7 (TX) of RFID1. Pin 10 of the board is connected to solenoid driver transistor T1 through base resistor R1. When pin 10 goes high, T1 conducts and solenoid is activated, which means lock is opened.

RFID reader module. In this project we used EM-18 RFID reader module (Fig. 2) operating at 125kHz. The mod-



Fig. 2: RFID reader module



Fig. 3: RFID tags

ule comes with an on-chip antenna and can be powered with a 5V power supply. The transmit pin (TX) of the module should be connected to receive pin (RX) of Arduino UNO board. Some specifications of the module are listed in Table I.

RFID tag. An RFID tag is a smooth card of credit-card size (Fig. 3), which is read by an RFID reader. It works at 125kHz and comes with a unique 32-bit ID. Normally, each tag has a unique ID number which cannot be changed. You can find out its unique ID through software.

Software

The software for this project is written in Arduino programming language. The Arduino UNO is programmed using Arduino IDE software. Atmega328

on Arduino UNO comes with a boot loader that allows you to upload new code to it without the use of external hardware programmer. It communicates using STK500 protocol. You can also bypass the boot loader and program the microcontroller through in-circuit serial programming (ICSP) header, but with boot loader the programming is quick and easy.

There are two sketch software codes for this application: ReadTag.ino for reading the tag's unique ID and AccessControl.ino for the main application.

Reading tag ID. To read the tag's unique ID, first you need to upload ReadTag.ino sketch into Arduino board by following the steps given below:

1. Connect TX pin of RFID1 to RX pin of Board1 as shown in the Fig. 1
2. Launch Arduino IDE. Connect Board1 to your PC and select the correct serial COM port and Board from Tools menu. Open ReadTag.ino sketch and compile it using Arduino IDE.
3. Burn ReadTag.ino sketch into the microcontroller in Arduino board
4. Open Serial Monitor from Tools menu in Arduino IDE
5. Hold RFID tag close to RFID1
6. Note down the tag ID shown on the Serial Monitor window. This ID will be used in the sketch later in the main application

Programming the main application. The main application of this project is access control system. So we need to upload AccessControl.ino sketch into the microcontroller in Board1. Note that before uploading the sketch into the microcontroller, you should remove the RFID reader mod-

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